

# Shan Yang

Staff Applied Scientist — Foundation Models, Multimodal AI, Generative AI

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## PROFESSIONAL SUMMARY

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- Technical leader with 8+ years driving foundation model research and production ML systems at scale across multimodal generative AI, video/image generation, and LLM post-training
- Proven track record managing Applied Scientists and establishing technical roadmaps, shipping products serving millions with \$500M+ revenue impact
- Deep expertise in multimodal foundation models: 3,500+ citations at premier venues (NeurIPS, ICCV, ECCV, ICML, AAAI), pioneering novel architectures in attention-based fusion, music-conditioned motion synthesis, and autoregressive diffusion

## TECHNICAL SKILLS

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**ML Frameworks:** PyTorch, TensorFlow, Scikit-learn — Distributed training (DeepSpeed, FSDP)

**Foundation Models:** Diffusion Models, Multimodal LLMs, Transformers, CLIP, BLIP

**Research Focus:** Novel Model Architectures, Image/Video/3D Generation & Editing, Multimodal Understanding, Post-Training (RLHF), Test-Time Training, Reinforcement Learning

**Architecture Innovation:** Attention Bottlenecks, Multimodal Fusion, Music-Conditioned Motion Synthesis, Autoregressive Diffusion, Linear Recurrent Models

## ENGINEERING & RESEARCH EXPERIENCE

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### Adobe Foundry

Staff Applied Scientist

San Jose, CA

March 2026 – Present

- Driving post-training research for Adobe Foundry, Adobe's family of generative AI foundation models powering Firefly and creative tooling at scale
- Developing post-training techniques (SFT, RLHF/DPO, LoRA, distillation, test-time training) to align foundation models with creator intent and production quality bars

### Amazon (Prime Video & Search)

Senior Applied Scientist Tech Lead

Palo Alto & Sunnyvale, CA

September 2021 – March 2026

- **Prime Video (2025–2026):** Led GenAI Live Action Studio with **5 Applied Scientists**, building autoregressive diffusion architectures and training infrastructure for 10B+ parameter video generation models; advanced controllable generation via novel post-training (Test-Time Training, Teacher-Diffusion-Forcing, RL)
- **Search (2021–2025):** Led **Amazon Video Search** from zero to launch serving millions with **\$500M+ annual revenue gain**, managing **3 Applied Scientists and 2 MLEs** and influencing a 15-person SDE team; pioneered multimodal fusion architecture integrating video, text, image, and structured data
- Drove multimodal LLM robustness research: Schema-of-Thought framework achieving **+28% hallucination reduction**; published at ECCV, AAAI and filed multiple patents

### Google Research

Senior Research Software Engineer

Mountain View, CA

June 2017 – September 2021

- Led foundational research in multimodal generative models, influencing **3 Research Engineers**, driving research agenda in music-conditioned 3D motion synthesis and video-audio-motion modeling, publishing at premier ML conferences (ICCV, NeurIPS, ECCV)
- Drove technical ownership of 2D/3D pose reconstruction models deployed on GCP at scale, demonstrating ability to lead end-to-end ML product development from research to production
- Provided technical leadership through patent contributions, cross-functional collaboration, and mentorship of research interns on advanced ML projects

## RESEARCH IN PROGRESS

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**Teacher Diffusion Forcing:** Novel training paradigm merging teacher forcing with diffusion forcing for autoregressive diffusion, modeling error as conditional distribution

## SELECTED PUBLICATIONS

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### Physics-R1: An Audited Olympiad Corpus and Recipe for Visual Physics Reasoning

**S Yang**

arXiv 2026 (2605.14040)

*Sole-author preprint. Three-stage corpus audit (Jaccard  $\rightarrow$  embedding  $\rightarrow$  LLM-judge), novel-source held-out olympiad benchmark (PhysOlym-A, 500q), and GSPO+DAPO recipe lifting Qwen3-VL-8B by +18.3 pp on PhysOlym-A.*

### Beyond Test-Time Training: Learning to Reason via Hardware-Efficient Optimal Control

P Wang, **S Yang**, X Wang, T Xiao, X Liu, C Yu, Y Lou, P Li, Z Wang, MC Lin, R Vidal

ICML 2026

*Linear recurrent architecture leveraging optimal control theory for LLM test-time reasoning (+35.8% Math-500, +10% AIME25).*

### Schema Perception for Robust Video Question Answering

X Wang, **S Yang**

Internal 2025

*Schema-of-Thought prompting framework for multimodal LLM robustness and OOD generalization (+28% Claude 3.5, +13.3% GPT-4o).*

### ViLA: Efficient Video-Language Alignment for Video Question Answering

X Wang, J Liang, CK Wang, K Deng, Y Lou, MC Lin, **S Yang**

ECCV 2024

*Efficient vision-language multimodal foundation models.*

### ICAR: Image-based Complementary Auto Reasoning

X Wang, A Liang, J Liang, M Lin, Y Lou, **S Yang**

AAAI 2024

*Multimodal reasoning for visual understanding tasks.*

### Attention Bottlenecks for Multimodal Fusion

A Nagrani, **S Yang**, A Arnab, A Jansen, C Schmid, C Sun

NeurIPS 2021 — **800+ citations**

*Multimodal fusion architecture for foundation models.*

### AI Choreographer: Music Conditioned 3D Dance Generation with AIST++

R Li\*, **S Yang\***, DA Ross, A Kanazawa

ICCV 2021 — **700+ citations**

*Multimodal generative modeling: music, motion, vision.*

### Physics-Inspired Garment Recovery from a Single-View Image

**S Yang**, Z Pan, T Amert, K Wang, L Yu, T Berg, MC Lin

ACM ToG 2018 — **100+ citations**

*3D generation from images.*

### Modeling Context in Referring Expressions

L Yu, P Poirson, **S Yang**, A Berg, T Berg

ECCV 2016 — **1600+ citations**

*Vision-language understanding.*

## PATENTS & INTELLECTUAL PROPERTY

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**V LAP: Efficient Video-Language Alignment via Frame Prompting and Temporal Distilling** (2023)

**Monitoring Animal Pose Dynamics from Monocular Images** (2022)

**Image-based Complementary Item Recommendation Algorithm** (2022)

**Auto Design: ML-based Furniture Layout Recommendation Algorithm** (2022)

## EDUCATION

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**University of North Carolina at Chapel Hill**

Ph.D., Computer Science

Chapel Hill, NC

2012 – 2017

**Shanghai Jiao Tong University**

B.E., Software Engineering

Shanghai, China

2008 – 2012